SPARC II

Shuffled-Radius Nulls, Halo-Catalogue Cross-Check Interface, and Morphology Splits

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2026

Abstract

Paper 61 introduced a first SPARC real-data pilot for MAAT structural dark-residual diagnostics. It showed that the structural diagnostic D_{struct} carries non-random galaxy-level signal relative to lightweight NFW-like and RAR/MOND-like comparison channels. This paper performs the predeclared robustness follow-up.

We test three upgrades. First, a shuffled-radius null destroys the radial pairing of $D_{\text{struct}}(r)$ within each galaxy while preserving each galaxy's distribution of structural scores. Second, galaxies are split into baryonic morphology proxies derived from the available SPARC rotation-curve mass components. Third, the code implements an optional published halo-catalogue cross-check interface for Li-et-al.-style SPARC halo-fit tables. No such local halo-catalogue table is redistributed in the present repository, so the catalogue cross-check is reported as not run in this execution.

The radius-null result is channel-dependent. The pooled within-galaxy radial correlation between D_{struct} and the NFW-like residual channel is

$$\rho_{\text{S}} = 0.3930,$$

far above the shuffled-radius 95% absolute null threshold

$$|\rho_{\text{S}}|_{95} = 0.0355.$$

By contrast, the RAR channel gives only

$$\rho_{\text{S}} = 0.0123,$$

below its shuffled-radius 95% threshold. Thus the Paper-61 signal survives the stronger radial null in the NFW-like channel but not in the RAR channel.

Morphology-proxy splits show that the mean structural residual is largest in dwarf-like and dark-matter-dominated disk proxies, while the strongest radial NFW-like alignment appears in bulge and ordinary disk proxies. The result is therefore not a universal dark-matter claim. It is a more precise statement: the MAAT structural residual carries radius-resolved SPARC information in some channels and galaxy populations, but the full published-halo-catalogue comparison remains future work until the external Li-et-al. table is supplied locally.

1 Purpose

Paper 61 was deliberately modest. It did not claim that MAAT explains dark matter, replaces cold dark matter, or outperforms professional halo modelling. It asked a smaller question:

> *Can the structural residual diagnostic be applied to real SPARC rotation curves, and does it carry non-random signal?*

Paper 65 asks a stricter question. If D_{struct} is meaningful, it should not merely survive galaxy-level shuffling. It should also retain signal when the radial ordering inside each galaxy is tested. Moreover, the signal should be inspected across galaxy populations and prepared for comparison with published SPARC halo-fit catalogues.

2 Inputs

The input data are the Paper-61 SPARC-derived radius diagnostics:

`paper61_radius_diagnostics.csv`, `paper61_galaxy_summary.csv`.

These are derived from the SPARC rotation-curve data of Lelli, McGaugh, and Schombert [1]. The present run contains

$$N_{\mathrm{gal}} = 175, \qquad N_{\mathrm{rad}} = 3391.$$

The SPARC input is used with the same fixed stellar mass-to-light factors as Paper 61,

$$\Upsilon_{\mathrm{disk}} = 0.5, \qquad \Upsilon_{\mathrm{bulge}} = 0.7.$$

This is still a simplifying assumption. Paper 65 is a robustness diagnostic, not a precision galaxy-fit analysis.

3 Shuffled-Radius Null

Paper 61 used a shuffled-galaxy null: galaxy-level values of D_{struct} were permuted against mismatch channels. Paper 65 adds a stronger radial null. For each galaxy g, the observed set

$$\{D_{\mathrm{struct}}(r_i)\}_{i\in g}$$

is preserved, but randomly reassigned to the radii of the same galaxy. This preserves:

- the galaxy identity;
- the distribution of D_{struct} values in that galaxy;
- the number and spacing of radial samples;
- the galaxy-level residual amplitude.

It destroys only the radial pairing between $D_{\mathrm{struct}}(r)$ and the comparison channels.

Before pooling, each quantity is standardised within each galaxy. The reported statistic is therefore a within-galaxy pooled Spearman correlation:

$$\rho_{\mathrm{S}}(z_g[D_{\mathrm{struct}}(r)], z_g[M(r)])\,,$$

where $M(r)$ is either the NFW-like fractional residual channel or the RAR/MOND-like sigma residual channel.

4 Halo-Catalogue Cross-Check Interface

Li et al. [6] provide a comprehensive catalogue of dark-matter halo models for SPARC galaxies, fitting seven halo profiles including pISO, Burkert, NFW, Einasto, DC14, coreNFW, and Lucky13. Those published fits are the correct next external baseline.

The Paper-65 code therefore accepts an optional local CSV file:

```
python3 sparc_ii_paper65.py --halo-catalog path/to/li_halo_catalog.csv
```

The expected minimal columns are:

`galaxy`, `chi2`

with flexible aliases for names and reduced-χ^2-like columns. For multi-model catalogues, the script keeps the best available fit per galaxy and reports the correlation between D_{struct} and the published halo-fit mismatch.

Status in this run. No local Li-et-al. halo-catalogue table is present in the repository. Therefore the published catalogue comparison is not executed here. The figure and JSON output explicitly mark this as

`not_run_no_local_catalogue`.

The internal NFW-like channel remains a lightweight proxy, not a professional halo-fit substitute.

5 Morphology and Proxy Splits

The prepared Paper-61 rotation-curve CSV does not include Hubble type. The code therefore supports two modes:

- if a local morphology catalogue is supplied, it is joined by galaxy name;
- otherwise, baryonic morphology proxies are derived from the available SPARC mass components.

  The fallback proxy classes are:

- `bulge_proxy`;
- `disk_proxy`;
- `dwarf_proxy`;
- `gas_rich_late_proxy`;
- `dm_dominated_disk_proxy`.

These are not official Hubble types. They are operational population splits based on maximum rotation speed, gas fraction proxy, bulge fraction proxy, and outer residual fraction.

6 Results

6.1 Radius-Shuffle Null

Channel	**Real** ρ_S	**Null** $\lvert\rho_S\rvert_{95}$	**Exceeds?**
NFW-like residual	0.3930	0.0355	yes
RAR residual	0.0123	0.0336	no
Radius itself	0.1241	0.0332	yes

Table 1: Within-galaxy pooled shuffled-radius null results.

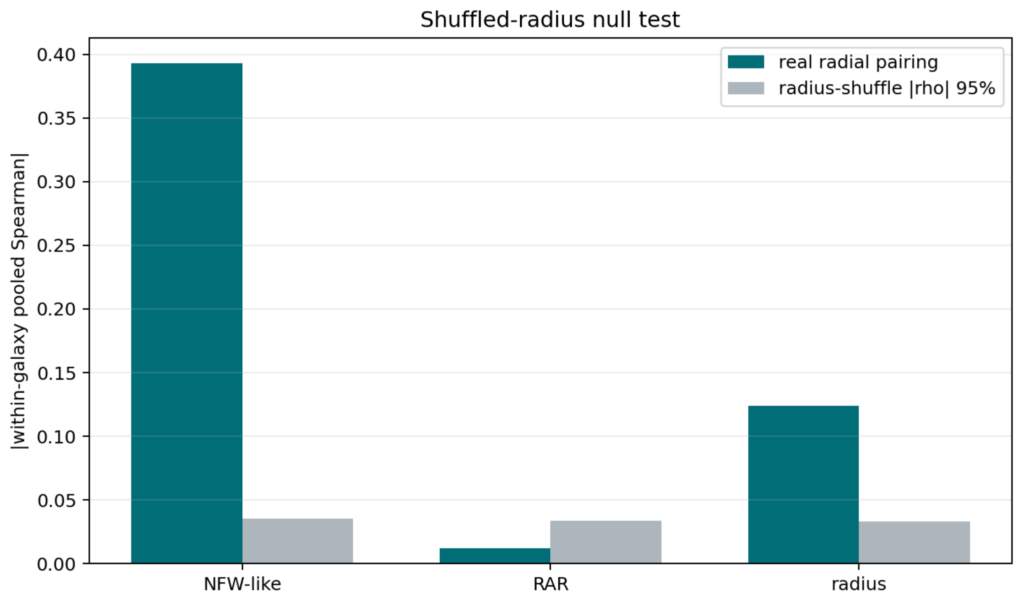

Figure 1: Shuffled-radius null test. The NFW-like channel remains far above the radius-shuffled null, while the RAR channel does not.

This is the central Paper-65 result. The structural diagnostic is not merely a galaxy-level pairing artifact in the NFW-like channel: its radial ordering carries information. But the result is not channel-universal. The RAR channel does not survive the same radial test.

6.2 Galaxywise Radial Correlations

For galaxies with at least five radial samples, the median galaxywise correlations are:

Quantity	**Median galaxywise** ρ_S
D_{struct} vs NFW-like residual	0.3000
D_{struct} vs RAR residual	-0.1227
D_{struct} vs radius	-0.0350

Table 2: Galaxywise radial correlation medians for 171 galaxies with at least five radial points.

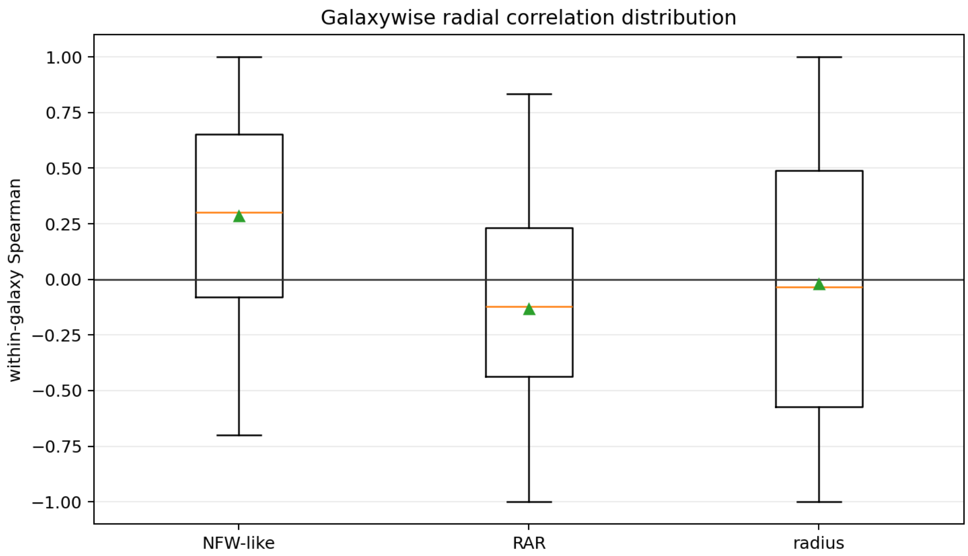

Figure 2: Distribution of galaxywise radial correlations.

6.3 Morphology-Proxy Splits

Proxy class	**Galaxies**	$\langle D_{\text{struct}} \rangle$	**NFW radial** ρ_S	**RAR radial** ρ_S
bulge proxy	26	0.0883	0.6473	0.2070
disk proxy	38	0.1218	0.4536	-0.0668
dark-matter-dominated disk proxy	32	0.2191	-0.0104	0.1564
dwarf proxy	70	0.2436	0.2402	-0.2372
gas-rich late proxy	9	0.2193	0.2394	0.1416

Table 3: Baryonic morphology-proxy splits. These are not official Hubble types; they are operational classes derived from the available mass-model components.

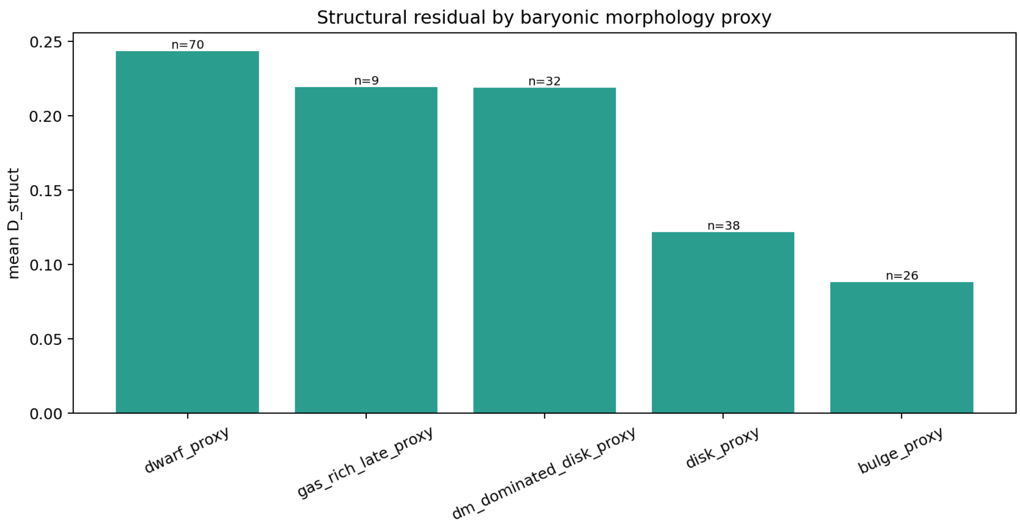

Figure 3: Mean structural residual by baryonic morphology proxy. Dwarf-like and dark-matter-dominated disk proxies have the largest average structural residual, while bulge/disk proxies show stronger radial NFW-like alignment.

6.4 Halo-Catalogue Slot

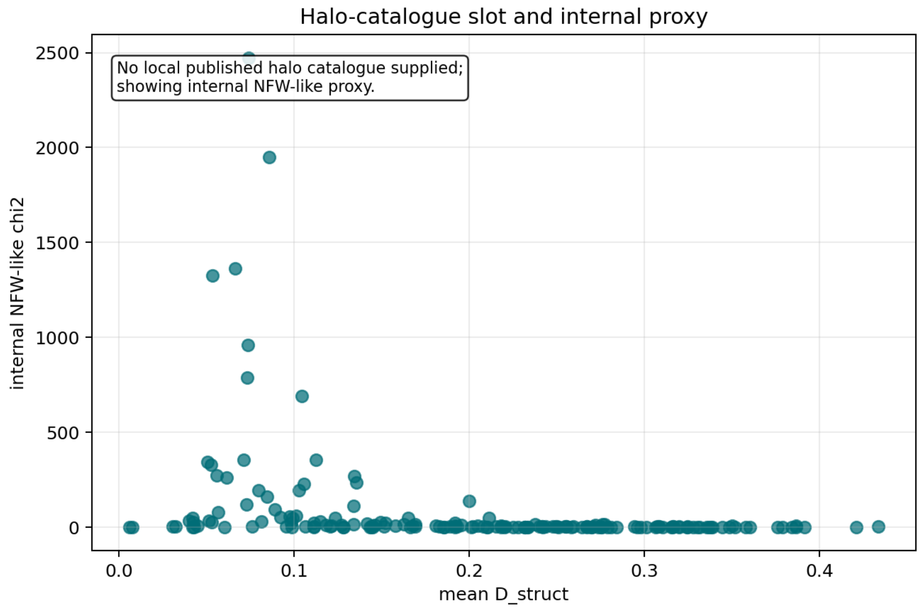

Figure 4: Published halo-catalogue cross-check slot. Since no local Li-et-al. table is supplied in this run, the plot shows the internal NFW-like proxy and marks the catalogue comparison as not yet executed.

7 Interpretation

Paper 65 strengthens Paper 61 in one important respect:

$$D_{\text{struct}}(r)$$

has radius-resolved information in the NFW-like channel that survives a within-galaxy shuffled-radius null. This is stronger than the Paper-61 shuffled-galaxy result because it preserves each galaxy's structural-score distribution and destroys only the radial assignment.

The result also narrows the claim. The RAR channel does not show the same radius-resolved signal. Morphology-proxy splits reveal that the diagnostic behaves differently across galaxy populations. The strongest average structural residual appears in dwarf-like and dark-matter-dominated disk proxies, while the strongest NFW-like radial alignment appears in bulge and disk proxies.

Thus the correct interpretation is:

> *SPARC structural residuals contain non-random radial information, but the signal is channel- and population-dependent.*

This is not a dark-matter solution. It is a robustness result for a structural diagnostic.

8 Limitations

- The published Li-et-al. halo-catalogue cross-check is implemented but not executed in this run because no local catalogue table is present.
- Morphology splits are proxy classes, not official Hubble types.

- Fixed stellar mass-to-light ratios from Paper 61 are retained.
- The NFW-like channel is a lightweight shape proxy, not a professional halo fit.
- The RAR channel fails the shuffled-radius null in this implementation.
- No lensing, CMB, abundance-matching, or full Bayesian halo inference is performed.

9 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle

Experiment folder:

experiments/sparc_ii_paper65/

Run:

```
cd experiments/sparc_ii_paper65
python3 sparc_ii_paper65.py
```

Optional catalogue run:

```
python3 sparc_ii_paper65.py --halo-catalog path/to/li_halo_catalog.csv
```

Outputs include:

- paper65_radius_shuffle_nulls.csv,
- paper65_radius_null_summary.json,
- paper65_galaxywise_radial_correlations.csv,
- paper65_morphology_proxy_splits.csv,
- paper65_galaxy_summary_with_proxy_morphology.csv,
- paper65_summary.json,
- four PNG figures.

Data attribution and license note. SPARC data are attributed to Lelli, McGaugh, and Schombert [1]. The local SPARC data used in the repository were prepared from the Zenodo-hosted SPARC rotation-curve record, DOI 10.5281/zenodo.16284118, listed under CC-BY-4.0. Derived CSV, JSON, and PNG files are reproducibility artifacts and must be distinguished from the original SPARC measurements. If a published halo catalogue such as Li et al. [6] is supplied locally, cite the original catalogue source and respect its distribution terms. No endorsement by SPARC, Li et al., Zenodo, VizieR, CDS, or any original data provider is implied.

10 Conclusion

Paper 65 performs the predeclared SPARC-II robustness step. The shuffled-radius null shows that D_{struct} retains strong radius-resolved alignment with the NFW-like residual channel. The same is not true for the RAR channel. Morphology-proxy splits show that the signal is population-dependent rather than universal.

The result is therefore useful but bounded: SPARC structural residual diagnostics now pass a stronger radial-null test in one channel, but they are not yet a published-halo-catalogue result. The next required step is to supply the Li-et-al. machine-readable halo table, repeat the analysis with official morphology information, and test whether the radius-resolved signal survives against professional multi-halo fits.

References

[1] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astronomical Journal* **152**, 157 (2016).

[2] J. F. Navarro, C. S. Frenk, and S. D. M. White, "The Structure of Cold Dark Matter Halos," *Astrophysical Journal* **462**, 563–575 (1996).

[3] J. F. Navarro, C. S. Frenk, and S. D. M. White, "A Universal Density Profile from Hierarchical Clustering," *Astrophysical Journal* **490**, 493–508 (1997).

[4] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," *Astrophysical Journal* **270**, 365–370 (1983).

[5] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters* **117**, 201101 (2016).

[6] P. Li, F. Lelli, S. McGaugh, J. Schombert, and M. S. Pawlowski, "A Comprehensive Catalog of Dark Matter Halo Models for SPARC Galaxies," *Astrophysical Journal Supplement Series* **247**, 31 (2020).

[7] C. Krieg, "SPARC Pilot Test of Structural Dark-Matter Residuals," preprint (2026).